Making Origami Digital Musical Instruments

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Abstract This paper presents an experimental study in post-digital luthiery using origami geometries. We explore how our method of fold sensing creates a new class of digital musical instruments through a case study of music and visual composition and performance. We consider how single-fold and multi-fold gestures afford an exploration of sound as a space, where each fold-sensed crease creates a new variable dimension in the musical parameter space, and multi-fold gestures allow multiple parameters to be controlled through origami gestures. We discuss our fabrication method, including material selection, sensor construction, and the selection

This research was funded in whole or in part by the Austrian Science Fund (FWF) [10.55776/AR590].

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G. Lu et al. (eds.), *Origami8, Volume II*, Lecture Notes in Mechanical Engineering,
https://doi.org/10.1007/978-981-96-8661-2_13

of origami patterns. The raw sensor data from the instrument are interpreted with our fold-sensing method that can infer four distinct but related types of gestural interaction, including touch, pressure, single-fold sensing, and multi-fold sensing. The data stream conforms to digital music standards, Musical Instrument Digital Interface (MIDI) and Open Sound Control (OSC). We introduce our open-source software toolkit, designed for fostering artistic experimentation, that implements and supports the musical standards. Our case study includes three distinct artistic strategies for interpreting the data stream and discusses the types of gestures employed by each artist in composition and live performance. We conclude that origami gestures show distinctly new types of expressions for both music and visual digital artists.

1 Introduction

Our work is situated in Art-Science Research, where our work is primarily led by an artistic aim. For this work, our artistic vision is to create a performable digital origami instrument, where folding and touching of the instrument generates a digital data stream for audio and or visual performance. We discuss the design and fabrication (making) method for creating Origami Digital Musical Instruments, also called *Oribotic*[1] *Instruments*, to generate a real-time stream of multimodal data into digital musical signals such as Open Sound Control or MIDI. We apply our related work in fold sensing, for technical information on fold sensing using Capacitance see our related paper *Fold Sensing Origami Gestures—A Case Study with Kresling Kinematics* [4]. This work extends previous engineering work from the field of origami robotics by exploring the question of fold sensing. In this case, we are motivated to explore folding as a mode of artistic and musical expression. Our oribotic instruments afford a novel experience: the haptic gesture of folding the origami structure and the digital sonification of folding a sound space. Our paper presents several contributions: the development of methods to use folding as a primary method to create digital music; a method for making capacitive-based fold sensing instruments that afford fold-gesture, touch and pressure interactions from one sensor type; and an evaluation of three artistic performances using oribotic instruments. The code for *oribotic instruments* is released as open-source and is published on GitHub.[2]

[1] **Oribotics**: ori (folding) + bot (robotics). First used as a term for a 2004 artwork titled *oribotics [origin]* https://matthewgardiner.net/art/oribotics_origin, discussed in [5, 6].

[2] https://github.com/oribotic/oribotic-instruments.

2 Related Work

2.1 Human Computer Interaction (HCI) and Musical Instruments

Digital and electronic musical instruments generally transform kinetic movement from the body, into a digital signal that is amplified or processed digitally and sonified or visualised in artistic performance. In [10], the authors reflect on Gibson's theory of affordances [8], where musical instruments create their interaction modes, keyboards afford key pressing, guitars afford strumming and in the case of Accordions, afford pushing and pulling. In the case of the electric guitar [19], playing the guitar causes string vibrations that are sensed by the electromagnetic 'pick-ups' and amplified. Playing an electrical guitar is directly analogue to playing an acoustic guitar, with some additional features such as rotational knobs for controlling the level of amplification. Many musical instrument interfaces use knobs and sliders that are based on potentiometers that allow for kinetic motion to transform the audio signal. In this work, we examine how *folding* can be used for musical expression. For example, folding is clear with the Accordion and Konzertina, where a folded bellows is used to pump air through the reeds of the instrument. Other examples where folding is part of performing a musical instrument are more difficult to find.

However, in the field of HCI, origami and the fold have received attention as a form of continuous input-output device. In [1] the authors present several foldable gadgets as haptic layers between electronics. Fold IO, [16] in addition to developing a toolkit-like method, uses a 10 Khz AC transmitter and receiver to calculate foldable angle for interactions to detect multiple types of shape-changing and foldable interactions. Commercial products such as the Bareconductive[3] components and conductive paint provide an open and makeable set of tools for creating experimental instrument interfaces, such as keyboards and other touch-sensitive interactions. Similar methods can be found in the capacitive interfaces using makeable methods such as printing and cutting in [15, 18]. In the case of oribotic instruments, the affordance of folding can be similar to push-pull interactions, but the folded pattern's geometry defines the instrument's gestures in space. The folds become a key mode for manipulating the sound space. A similar theme is found in [9], where the authors use capacitive sensing between parallel plates to sense the geometric change in auxetic structures, translating compliant hinged mechanisms to origami structures. Finally, there is extensive work in HCI and new musical interfaces.[4] These works present different aspects of the fold as Input Output (IO), often using paper, charmingly as their medium. While our work began with and explored paper prototypes, due to the time consumption required to fabricate oribotic instruments, our attention went towards materials that could withstand folding and unfolding interactions for extended periods and be used in performances or exhibitions.

[3] https://www.bareconductive.com/.

[4] See New Interfaces for Musical Expression https://www.nime.org/.

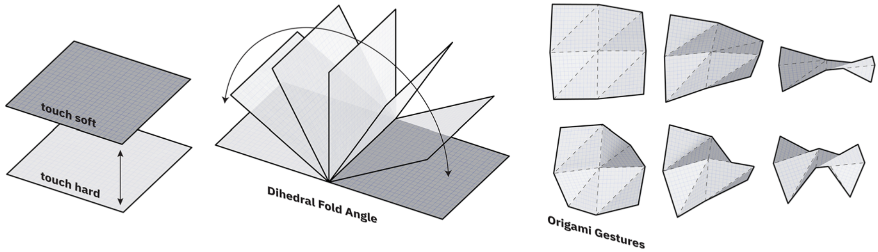

Fig. 1 Touch, fold-Sense and origami gestures. Our artistic concept was to create an instrument that embodies origami affordances where touch and folding are the primary haptic interactions

3 Method

3.1 Design Considerations

We aimed to create a device to respond to fold and touch (see Fig. 1). At the simplest level, the origami gesture is the continuous range of motion of the Dihedral Fold Angle (DFA), measured along the fold line between two faces. At the more complex level, the origami gesture can be estimated through the sensing of multiple folds in the structure. To achieve this Capacity Proximity Sensing (CPS), and Capacitive Touch Sensing (CTS), provide potential for our multimodal aim with a single sensor design. Additionally, capacitive sensing only requires one cable per panel, simplifying the manufacturing process and enhancing the dimension and pattern possibilities. CPS to determine the proximity between adjacent panels could be used to predict the DFA and CTS where a human finger touches the panel. Both sensing types have different optimum design parameters, the basics of which can be found in [2]. For our prototypes, we selected one design strategy that overlaps with CPS and CTS, a planar electrode that affords touch and proximity sensing.

3.2 Prototyping Process

Our prototyping process spans many iterations focused on evaluating the process of making, aesthetic and functional results. We began with paper, copper foils, and conductive threads such as Madiera HC40.[5] These afforded capacitive touch sensing quite effectively. However, interference resulted from adjacent panels and touching from the reverse side. We experimented with 3D-printed electrodes from a conductive graphene polymer and PCB electrodes over embroidered textile wiring, but the

[5] Madeira HC 40 is heavy, fully silver-plated polyamide embroidery thread with resistance of less than 300 Ω/m https://www.madeirausa.com/hc-40-sp-highly-conductive-embroidery-thread-40.html.

conductive thread suffered breakage over the fold lines. We then progressed to hand-soldered wires connecting the electrodes. During the process, we experimented with different grounding materials to shield the wiring to prevent signal interference. For grounding, our experiments functioned with inexpensive adhesive copper tape, but copper has significant fold memory and the potential to break after repeated foldings. To overcome this, we experimented with conductive textiles, they are robust and shield effectively.

3.3 Origami Geometry

We have experimented with several origami geometries in this work, see Fig. 2. In this paper, we present information focused on the Yoshimura variant.

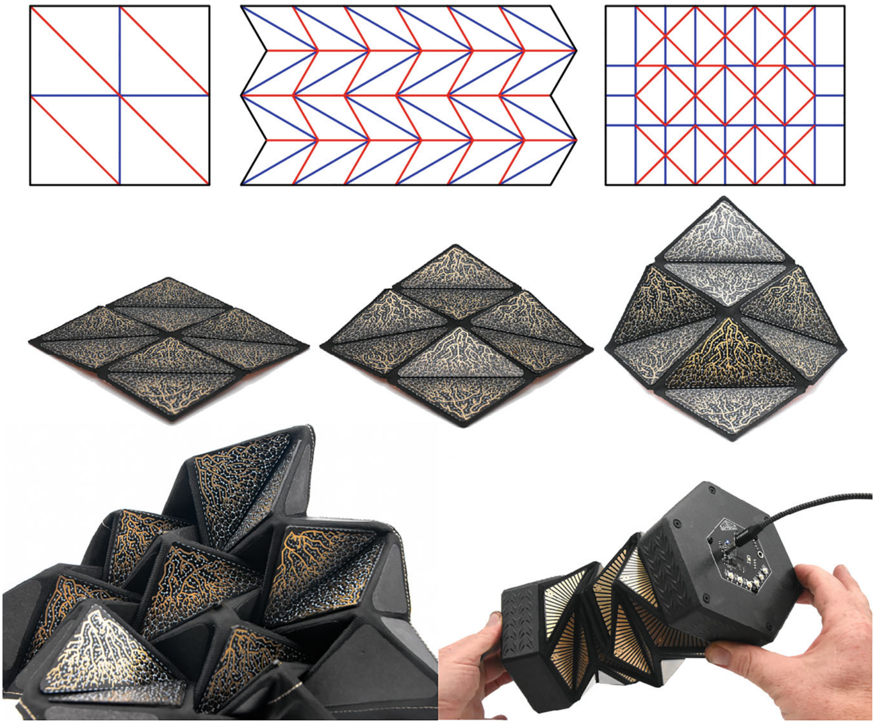

Fig. 2 Crease patterns and oribotic instruments. Top left: yoshimura, kresling, waterbomb. Middle row: Y8 yoshimura 8-electrode instrument. Bottom row: on left, a waterbomb-based geometry; on right, a kresling-based geometry, called the 'oricordion' after its visual similarity to the accordion instrument

1. A Yoshimura [23] pattern, called the *Y8*, is the simplest of the current family of instruments and affords simple interactions such as simultaneous compression and expansion of the entire structure and actuation of individual panel pairs.
2. A Kresling tube [11, 12], we call the *Oricordion*. The meta-mechanical origami properties of the Kresling Pattern, in particular this neutral rotation variation, is that it can twist, rotate and compress in multiple ways, in addition to the actuation of individual panel pairs. The origami-gestural affordances are exemplified in this instrument design. This design has 48 electrode panels.
3. A Waterbomb Pattern [12, 21], a design we call *Suki* that has 12 panels and uses Fold Printed [3] techniques to make other sections of the origami structure rigid in addition to the PCB electrodes.

3.4 Materials

In this section, we *briefly* discuss the materials tested, our basic criteria and our selection of materials. Each topic could be extended across several papers and appendices, but for the sake of brevity, we focus on our findings. The topics are the textile base, grounding textiles, bonding, electrodes, and hinging of wiring. **Textile Base**: in our prior work, including Oribotics [7] and Fold Printing [3], we considered the problem of durability of a foldable hinge for origami robotics. Textiles affording a durable hinge between two rigid or semi-rigid plates. We created successful prototypes with a variety of natural textile fibres, including felts, woven wool blends, woven cotton blends, and vinyl-coated weaves. For fast prototypes, we also recommend using paper as a surface for low-cost prototyping (see Fig. 3). **Grounding Textiles**: we tested the complete product lineup from Shieldex,[6] and based on their laser-cutability, the effect on the sensor grounding, flexibility, haptic and visual quality, we selected the following two textiles: *Shieldex®Prag*[7]: a woven polyester fabric metallised with 49% pure copper content. Laser cuttable, non-flexible with shielding properties. We used Prag between the non-conductive textile and wiring; and *Shieldex®Technik-tex P130 + B*[8]: A knitted textile, metallised with silver and copper coatings, which consists of 22% elastomer, making it stretchable in all directions. It is laser cuttable, with slight discolouration at the edges. We used this textile due to its flexibility and tactile qualities on the outer layer. **Bonding**: We use 3M Thermal Bonding Film 583[9] for all mechanical bonds. This film is dielectric and creates a reliable and durable bond to FR4. According to the 3M datasheet the 3M film holds at 600psi for FR4, but is untested on textiles. We have not mechanically tested the bond with each textile; however, the bond was sufficient for our prototypes and very useful for bonding

[6] https://shieldex.de/.

[7] https://www.shieldex.de/en/products/shieldex-prag/.

[8] https://www.shieldex.de/en/products/shieldex-technik-tex-p130-b/.

[9] https://multimedia.3m.com/mws/media/169553O/3m-thermal-bonding-film-583.pdf.

flat materials with a heat press. An alternative method is craft glue suitable for textiles. It laser cuts cleanly.[10] **Capacitive Electrode Panels**: We tested FR4 PCBs[11] with thicknesses 0.4mm to 1.6mm with different platings (ENIG),[12] HASL[13] and Lead-Free HASL. The rigidity of the PCB restricts movements of the geometry to the kinematics of the origami pattern. This technique was popularised as a fashion accessory by Issey Miyake in his BaoBao Bag series [14], where rigid polymer plates are glued onto a flexible plastic substrate. Our design currently has three main features. (see Figs. 3 and 5). A solid electrode on the top and bottom of the PCB surface. We experimented with the top as an electrode and the bottom as ground; however, the capacitive signal range declined. Solderable pads are on the bottom of the board. The geometry of the electrode can be variable, and according to [2], the shape does not significantly affect proximity sensing, however, we found in the case of rotational or fold-angle sensing, the geometry does have an effect. The characterisation of electrodes is not discussed in this paper. **Hinging**: There is a longevity problem with folding and unfolding any wire. The ductile properties of copper mean that movements cause gradual stress that leads to breakage. To minimise this effect, we apply a method learned from flexible origami robotic design [13] known as the Tomoe-Type hinge [20], an S-bend was introduced to the wiring structure as it crosses the fold. This does not solve the longevity problem, but it does significantly improve longevity of the prototypes. The problem of wiring is an area for futher work.

3.5 Construction Layers

The construction is diagrammatically explained in Fig. 4. Each layer is laser cut and heat bonded. Wires are first laid out on a 3D-printed wiring jig, soldered by hand and further fixed with adhesives. The order of assembly is top to bottom.

3.6 Electronics

During our iterative prototyping, we found three critical factors: **electrode design**, **wiring** and **grounding**. The *electrode* can be fabricated using PCBs, affording rigidity, and increased design potential in terms of geometry, patterning and ease of fabrication for multiple electrodes. *Wiring* is more robust and signal quality is more

[10] Safety note. Laser cut only with proper air filtration and ventilation! 3M Thermal Bonding Film 583 is 0.05 mm thick, based on Nitrile Phenolic. Phenolic resins release formaldehyde during combustion. Phenolic resins are used in bonding plywood a common material for laser-cutting.

[11] FR4 Flame Retardant fibreglass composite, a class of Printed Circuit Board (PCB).

[12] Electroless Nickel Immersion Gold (ENIG).

[13] Hot Air Solder Levelling (HASL).

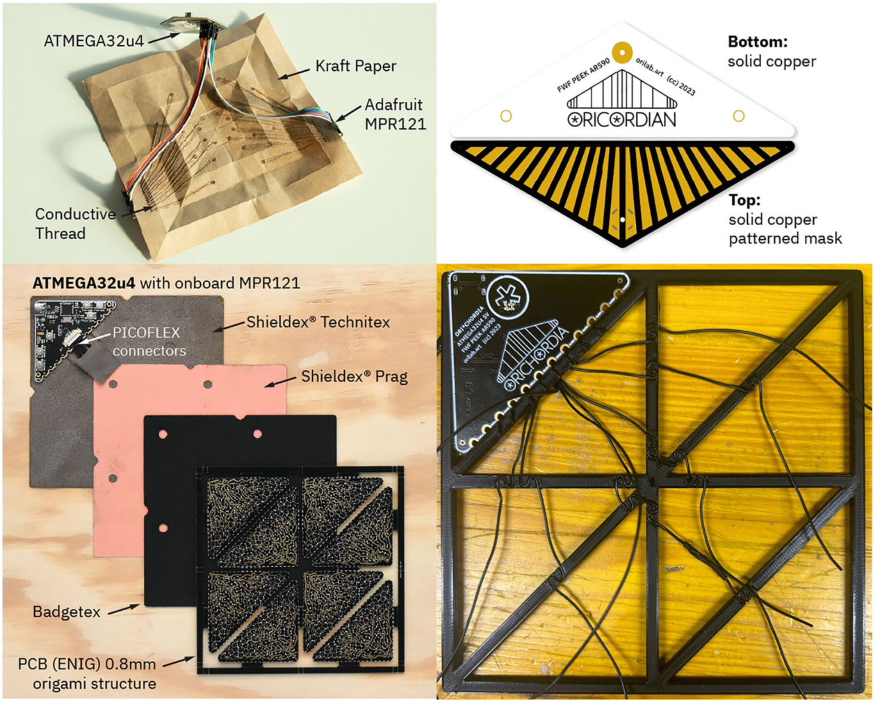

Fig. 3 Prototypes and materials. Top left: reverse side of an early (2021) paper and foil prototype with conductive yarn. Top right: an electrode design for the Kresling-ori. Bottom left: cut and prepared materials, see also Fig. 5. Note the PCB electrodes are fabricated within a frame with a 6mm fold gap, saving significant time with assembly. Bottom right: 3D printed wiring jig, allows for more precise and reliable wiring layouts

stable with hand-soldered wiring when compared to our earlier experiments with digital embroidery. However, the wiring jig for placing the cable needs careful design consideration over the folds see our comments on **hinging** above. Without *grounding* over the back of the instrument surface, aadjacent panels influence each other; with grounding, interference is minimised. The following section outlines electronic components.

Our prototypes examined the following electronic components. *Microcontroller*: The Y8 instrument has an ATMEGA32u4 an 8-bit AVR®RISC-based microcontroller used in the Arduino Micro, which has built-in USB interfaces, often used in custom keyboard designs, enabling the prototype as a USB-MIDI interface. *Capacitive Sensing*: MPR121 is a well-known capacitive interface with 12 capacitive interfaces. Our early prototypes used an off-the-shelf Adafruit 12-Key Capacitive Touch Sensor Breakout—MPR121.[14] Later custom circuit designs included MPR121 chips hard-wired I2c addresses for up to four MPR121 capacitive sensor chips allowing

[14] https://www.adafruit.com/product/1982.

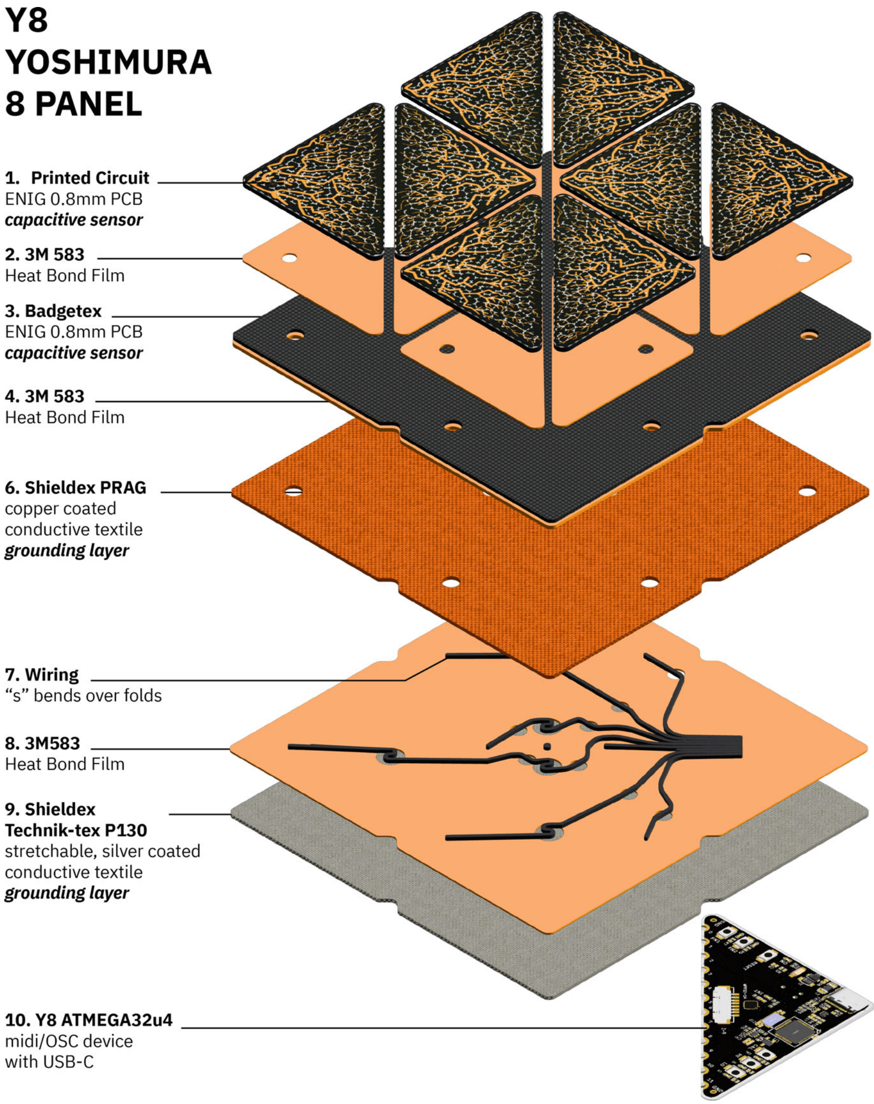

Fig. 4 Y8 construction

up to 48 capacitive panels. Our latest circuitry uses the RP2040 Raspberry Pi microcontroller with breakouts for two I2C channels, extending the system to 96 electrodes. *Ground Plane*: the textile layer, discussed above, that covers the entire reverse side of the interface to minimise electromagnetic interference between adjacent electrode pairs and the human hand during performance.

3.7 Software

Our software stack has three main layers: *firmware* to handle the digital signals between the capacitive sensor panels and the USB port of a computer; a *software* layer designed to simplify acquisition of the data stream via OSC or MIDI; and *creative software* to transform the data stream into audio or visual signals. We used Pure Data [17] for data flow programming for handling MIDI and OSC signals as the target platform for creative software development. However, any software that can handle either MIDI or OSC remains open for the artist. Artistic expression occurs on the creative software layer. This top level layer allows for linking gestures to artistic outputs such as music or visualisations, as examples, we discuss three different artistic approaches in the results.

3.7.1 Software Stack

1. **Oribotic Instrument Firmware**[15] (C++) specific to our custom circuit board hardware. Programmed with Arduino IDE,[16] an established and accessible framework for microcontroller programming. Including the following libraries: Bare Conductive MPR121 Arduino Library,[17] MIDIUSB,[18] a standard Arduino Library for MIDI interfacing via USB, Open Sound Control Arduino Library for OSC via USB.[19]
2. **Oribotic Instrument Software*** (Pure Data Abstractions) handling OSC or MIDI data for post-processing directly in Pure Data or to forward MIDI to a software or hardware synth and is included in the Github repository.
3. **Example instruments and analytic tools*** (Pure Data Abstractions and Python Code) including several examples of digital instruments and analytic tools, as starting points for artistic experimentation and data analysis.

3.8 Programming Logic of Sensor Interactions

Using the MPR121's touch register and filtered and baseline data from the MPR121, we identified four unique gestural affordances from these instruments' electrodes.

1. **Touch and Release**: the MPR121 is designed to identify touch and release interactions with each electrode. As a boolean value, these signals are useful for triggering samples or notes.

[15] * https://github.com/oribotic/oribotic-instruments/.
[16] https://www.arduino.cc/en/software.
[17] https://github.com/BareConductive/mpr121.
[18] https://www.arduino.cc/reference/en/libraries/midiusb/.
[19] https://github.com/CNMAT/OSC.

2. **Soft Touch**: examining the filtered data following a touch and before a release affords a calculation of pressure on the sensor, as increased or decreased pressure is measurable in small amounts. As an integer value with a low range, minor adjustments to volume on a sample, i.e. press harder = volume up, press lighter = volume down.
3. **Fold Sensing**: is a continuous data stream of the filtered data. The data indicates proximity to another electrode, human, or other grounded electromagnetic field. The sensor does not indicate the nature of the object, only proximity through change in the capacitive field. The instrument is very sensitive and responsive. Between electrode pairs, across the crease line, we can estimate an angle between adjacent electrodes using a characterisation of the sensor behaviour. The value is generally scaled to 0–254 for OSC or 0–127 for MIDI and can be used for controlling multiple sound or visual parameters with one folding gesture. For example, mapping the folding angle to the volume of a sound sample creates a folding amplification controller. More complex applications, such as mapping one or more folds to potential multi-parametric audio operations, became part of the artistic inquiry.
4. **Fold Gesture Sensing**: this topic is discussed in detail in *Fold Sensing Origami Gestures—A Case Study with Kresling Kinematics*. [Gardiner et al., 2024]. Briefly, this feature evaluates multiple folding angles for multiple folds, giving an aggregate estimate of a given origami gesture.

4 Results

We presented the Y8 oribotic instrument at the Ars Electronica Festival 2023 in two formats, as a performance of three musical works by three different artists, and secondly, as an interactive exhibition for the general public to experience our oribotic instruments hands-on. The brief to the artists was to create a short, maximum ninety-second performance. The notion was declared as *post-origami-punk* at the start of the performance. For the exhibition, the audience could switch between the compositions created by the three artists.

4.1 Performances

The following is a brief description of the compositions for the world premiere of *oribotic instruments*. See footnote[20] for video documentation.

[20] Video of the performance https://www.youtube.com/watch?v=MlSFuYIOzVc.

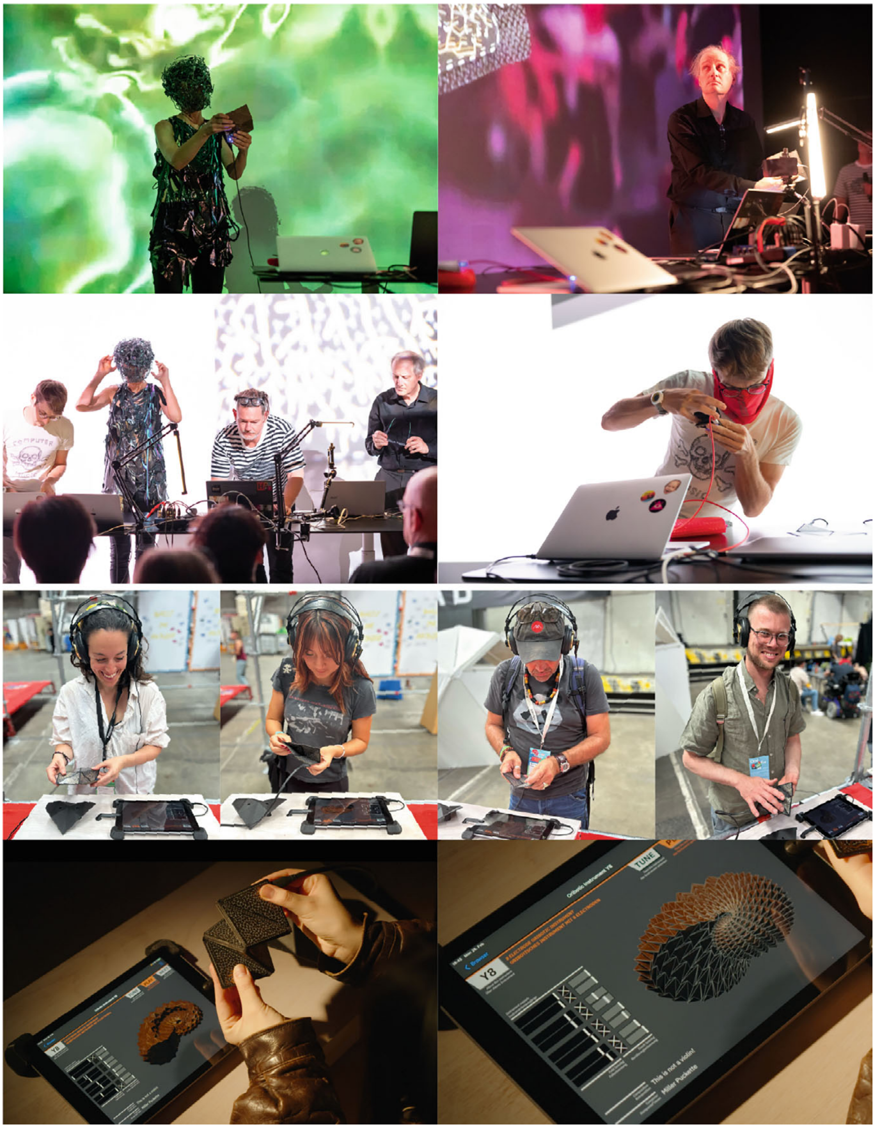

Fig. 5 Oribotic instruments performance and exhibition. Top: left Anne Wichmann folding ambient sound, right Miller Puckette amid simultaneous performance and observation. Second Row: from left Dan Wilcox, Anne Wichmann, Matthew Gardiner and Miller Puckette; right Dan Wilcox folding a wild solo. Third Row: characteristic audience interactions at the Ars Electronica Festival. Bottom: left video still showing the Y8 instrument and the user interface; right close up of interface. Photos of Performance in Ars Electronica's Deep Space 8K by Tom Mesic courtesy of Ars Electronica; Photos of exhibition by Matthew Gardiner

4.1.1 Anne Wichmann (She's Excited!): Sui-Ocean

In her performance, Anne Wichmann (She's Excited!) presented the piece "Sui-Rain," folding a multi-parameter path and navigating a range of origami-like gestures through eight different pre-composed audio-samples that would together create a multi-layered composition. Depending on the origami gestures, the soundscape would morph and change. Pre-recorded video sequences emphasised the ambient, fluid sonic space of water, rain, and the ocean. In discussions with Anne, the process of composition was described as being inspired to take the Y8 instrument through all of its various folding gestures, in a choreographic folded sequence that became the performance. This echoed another aspect of the artistic brief, which was to think of composition as a folded musical space.

4.1.2 Miller Puckette: This Is Not a Violin

The next performance came from Miller Puckette, the author of the graphical development environment Max/MSP and the real-time performing platform Pure Data. The title "This is not a Violin", referencing Henri Magritte's series of works, *The Treachery of Images*, is taken through Puckette's composition this is literally *not* a violin. The piece takes the audience into the chaotic world of noise-trending audio synthesis, inputting raw fold-sensing data from the instrument into a noise generator. Noisy, loud, experimental, is this origami-punk? Mixing folding, touch and open-ended chaos. The video accompaniment via live close-up camera feed brought the audience up close to the subtle gestural and responsiveness of the Y8.

4.1.3 Dan Wilcox (Robotcowboy): Noble Brass

As the main developer of the Pure Data abstractions, Dan arguably spent the most time experimenting with the Y8. In the piece titled "Noble Brass," he leverages every feature of the instrument. The performance began with a short digital touch-led percussion intro which alternated between two single tom panels and a full hand, 4 panel bass drum trigger. This lead into a melodic synth-pop exploration of soft touch Frequency Modulated brass voices, exploiting the touchable qualities of the instrument through trigger and velocity control along an ascending scale matching the natural layout of the panels. The transition to the second movement is a fluttering of light touches before descending into a squealing, fold-sensing guitar-solo-esque crescendo of cross frequency modulation. Dan's visuals are signal-driven, using a minimal pixel block interface, he establishes a live visible link between the instrument and data.

4.2 Exhibition

The exhibition[21] was designed to make oribotic instruments performable for the general public. To remove as many obstacles to the experience as possible. We used iPads as the computational device, made with PdParty, an open-source iOS application for running Pure Data patches on Apple mobile devices using libpd [22]. Users approach the exhibit, put on headphones pick up the instrument and instantly hear the generated audio. They could switch between compositions and also tune the instrument by following a three-step on-screen process. In the background, an abstract folded geometry shows a real-time visualisation of the overall folded state of the instrument. In the experience, the audience is engaged with three senses, haptically with their hands and fingers, visually from the on-screen display and primarily with hearing. The multimodal experience is highly engaging and we consistently observed extended usage sessions from audience members.

5 Discussion

The performances were openly experimental. The artists were given an open brief to extend their musical ideas. The results are a diverse range of musical expressions and interactions with our Y8 oribotic instrument design. This diversity comes from a relatively simple oribotic instrument: the Y8 instrument has eight panels and eight folds, and only four paired folds. Anne Wichmann's pensive and paced choreographic approach to folding her sound space fitted very closely with our original conception. That through folding the instrument, into different states, the performer could navigate a sound space. Miller Puckette evokes a chaotic crumpling and crushing of the instrument, something completely unexpected. We did not even imagine how this could be translated, and yet it does, as Puckette's hand gestures reinforce how one might hold and crumple a sheet of paper. The robotcowboy performance shows how the Y8 can be like a keyboard and create melody, and then become something completely different, a dynamic object you can hold and shake, or rapidly fold and unfold a resonant chaotic modulation effect.

The decision to proceed with the Y8 for the performance was not ours entirely. While the Y8 was finished first, and distributed to the artists in advance, they were allowed to experiment with the waterbomb and kresling-type instruments. For the general public visiting our exhibition, the minimalist design was really necessary. As with many media-art exhibits featuring new technology and completely novel forms of interaction, the user is often left with the rhetorical question, "*So what am I supposed to do with this?*". The exhibition instructions were deliberately minimal, we didn't explain how to use it, rather, we observed how it was used intuitively. We observed a diverse range of experimentation. Some audience members didn't comprehend that it folded, and got immersed in simply exploring the touch sensing,

[21] See https://orilab.art/ori/Oribotic_Instruments.

others found their way into the folded gesture immediately. A small percentage didn't even try to pick it up. Another observation was that some people came again and again, spending considerable time performing for themselves or others.

6 Conclusion

Our work with oribotic instruments opens many new artistic questions for composition and musical expression. Composition with oribotic instruments is not yet intuitive and simple. While our firmware and software make connecting and getting data from the instrument plug and play with MIDI, the composition method relies on programming knowledge in Pure Data or similar programming logic. For seasoned digital musicians our methods presents incredible flexibility and potential, but for a beginner audience, this presents a far higher learning curve of a challenge than plugging in a MIDI keyboard. But as Miller Puckette's title points out, *this is not a violin*, oribotic instruments are something else entirely. Additionally, there's no notation. Could origami diagrams be a form of notation for describing folding gestures and timing? On the other hand, performing or interacting with an oribotic instrument is as intuitive as folding. Our paper discussed several outcomes, including processes for building and making digital origami instruments, and presents a preliminary body of fundamental knowledge to facilitate further development and experimentation. This includes material selections made after extended prototyping, electronics and grounding solutions, supported by a release of open-source firmware and software. Our work shows that origami concepts, folding and unfolding, and more complex ideas like multi-fold gestures can be used to navigate a sound space. The performance results show a diverse range of new artistic directions are possible with folding sound. Oribotic instruments open up a new area of musical expression, composition and performance through folding.

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